

EE4C5/5C5 Digital Signal Processing

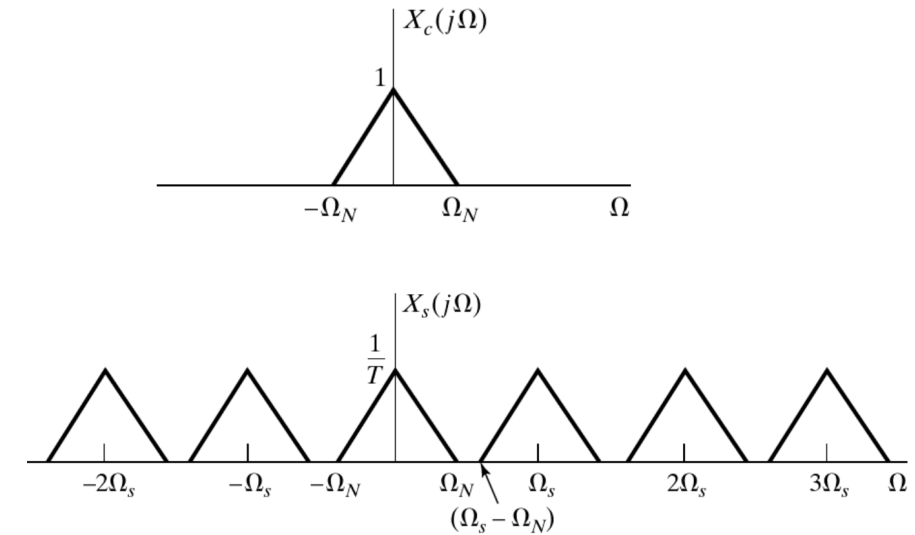
Lecture 5 – Decimation and Interpolation

This lecture

- Based on Chapter 4 of O&S
 - Section 6 onwards
- All images from O&S book unless otherwise stated

Recall...

- The impulse train approximation $x_s(t)$ of the original continuous time signal $x_c(t)$ has (continuous) Fourier Transform:
- $$X_s(j\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - k\Omega_s))$$
- With T the sampling period, Ω is the continuous-time frequency in radians/s, and $\Omega_s (= 2\pi f_s)$ is the angular sampling frequency in radians/s
- Saw that $X_s(j\Omega)$ comprises copies of $X_c(j\Omega)$ shifted by $k\Omega_s$ and scaled by $\frac{1}{T}$



Pay attention – discrete or continuous FT

- Can show that (how?)
 - $X_s(j\Omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega nT}$
- The DTFT gives
 - $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$
- So:
 - $X_s(j\Omega) = X(e^{j\omega}) \Big|_{\omega=\Omega T} = X(e^{j\Omega T})$
- Thus the DTFT $X(e^{j\omega})$ is a frequency scaled version of $X_s(j\Omega)$, with the scaling specified by $\omega = \Omega T$
- The above relates the CTFT to the DTFT

Useful for later...

- Combining we get:
- $X(e^{j\Omega T}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - k\Omega_s))$
- Or equivalently
- $X(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{\omega}{T} - \frac{2\pi k}{T}\right)\right)$

Resampling

Resampling a discrete-time sequence

- A continuous time signal $x_c(t)$ can be represented by the discrete sequence $x[n]$ where
 - $x[n] = x_c(nT), \quad -\infty < n < \infty$
- Often wish to change the sampling rate of $x[n]$ such that
 - $x_1[n] = x_c(nT_1), \quad T_1 \neq T$
- This is resampling
- Could we reconstruct original $x_c(t)$ and use new period T_1 ?
 - Not practical
 - Restrict to discrete time operations

Terminology reminder

- Ω_N the Nyquist Frequency
- $2\Omega_N$ the Nyquist Rate

Reduce Sampling Rate

Reduce sampling rate by integer factor M

- $x_d[n] = x[nM] = x_c(nMT)$
- With $X_c(j\Omega)$ bandlimited to $\pm\Omega_N$ then $x_d[n]$ will be exact representation if $\pi/T_d \geq \Omega_N$ or equivalently $\pi/MT \geq \Omega_N$
- Can avoid aliasing if:
 - Original sampling rate is at least M times the Nyquist rate
 - Bandwidth of the sequence is first reduced by factor M by discrete-time filtering

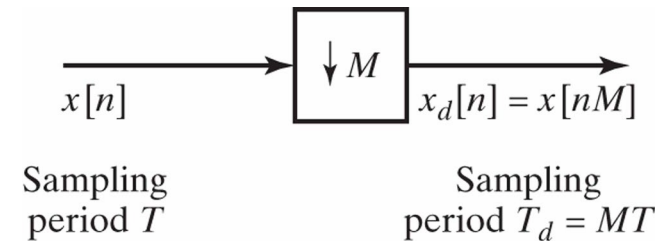
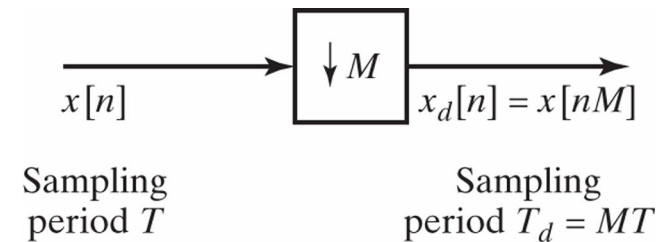


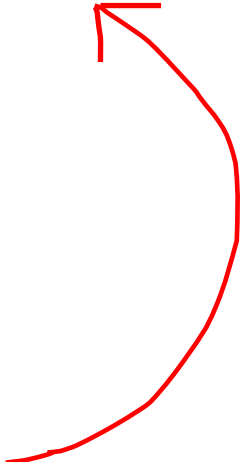
Figure 4.19 Representation of a compressor or discrete-time sampler.

Reduce sampling rate by integer factor M

- Using $X(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c \left(j \left(\frac{\omega}{T} - \frac{2\pi k}{T} \right) \right)$ (slide #5)
- DTFT of $x_d[n] = x[nM] = x_c(nMT)$
 - $X_d(e^{j\omega}) = \frac{1}{T_d} \sum_{r=-\infty}^{\infty} X_c \left(j \left(\frac{\omega}{T_d} - \frac{2\pi r}{T_d} \right) \right)$
- But with $T_d = MT$ can rewrite:
 - $X_d(e^{j\omega}) = \frac{1}{MT} \sum_{r=-\infty}^{\infty} X_c \left(j \left(\frac{\omega}{MT} - \frac{2\pi r}{MT} \right) \right)$



Reduce sampling rate by integer factor M

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 - DTFT of $x_d[n] = x[nM] = x_c(nMT)$
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 - But with $T_d = MT$ can rewrite:
 - $X_d(e^{j\omega}) = \frac{1}{MT} \sum_{r=-\infty}^{\infty} X_c \left(j \left(\frac{\omega}{MT} - \frac{2\pi r}{MT} \right) \right)$
 - The summation index r is related as
 - $r = i + kM$
 - k and i integers, $-\infty < k < \infty$ and $0 \leq i \leq M - 1$
- 
- Want to relate back

Reduce sampling rate by integer factor M

- $X_d(e^{j\omega}) = \frac{1}{M} \sum_{i=0}^{M-1} \left\{ \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c \left(j \left(\frac{\omega}{MT} - \frac{2\pi k}{T} - \frac{2\pi i}{MT} \right) \right) \right\}$
- Recognising that
 - $X(e^{j(\omega-2\pi i/M)}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c \left(j \left(\frac{\omega-2\pi i}{MT} - \frac{2\pi k}{T} \right) \right)$
- Can arrive at:
 - $X_d(e^{j\omega}) = \frac{1}{M} \sum_{i=0}^{M-1} X(e^{j(\omega-2\pi i/M)})$
- Expresses FT of $x_d[n]$ in terms of FT of $x[n]$
- $X_d(e^{j\omega})$ consists of M amplitude-scaled copies of $X(e^{j\omega})$, frequency scaled by M , and shifted by integer multiples of 2π .

Avoid aliasing...

- $X_d(e^{j\omega})$ periodic with period 2π (like all DTFT!!)
- Can avoid aliasing if $X(e^{j\omega})$ is bandlimited, i.e.
- $X(e^{j\omega}) = 0 \quad \omega_N \leq |\omega| \leq \pi$
- And $2\pi/M \geq 2\omega_N$
- In this example, original sampling rate is $2\pi/T = 4\omega_N$
- Thus with $M=2$, at the limit to avoid aliasing here.

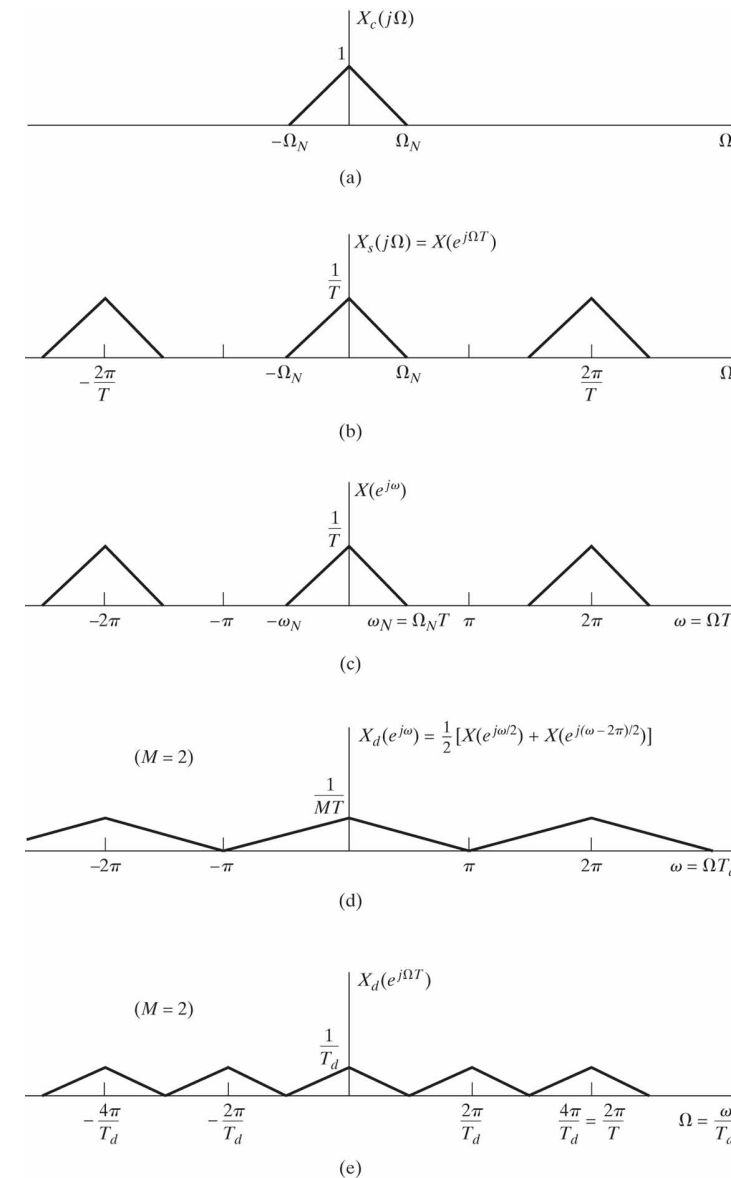


Figure 4.20 Frequency-domain illustration of downsampling.

Aliasing with M=3

- Will get aliasing with M=3
- Prefilter to avoid this and bandlimit the signal

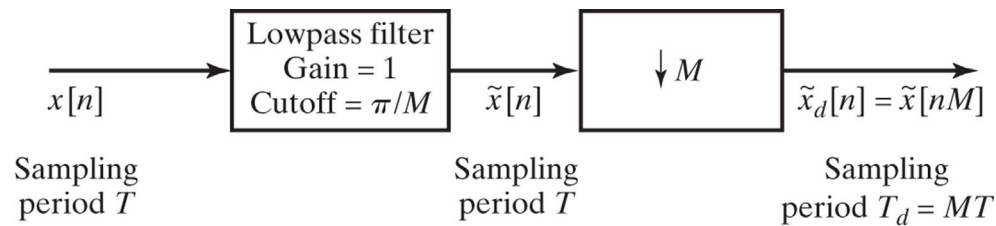


Figure 4.22 General system for sampling rate reduction by M .

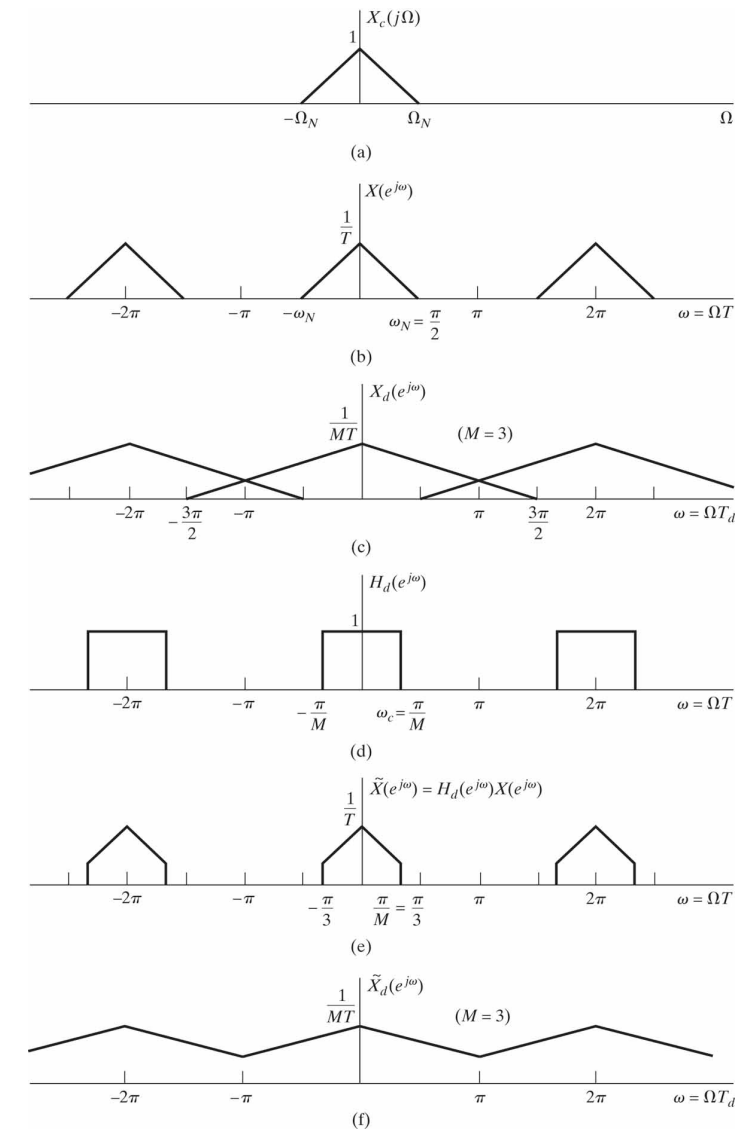


Figure 4.21 (a)–(c) Downsampling with aliasing. (d)–(f) Downsampling with prefiltering to avoid aliasing.

General Rule

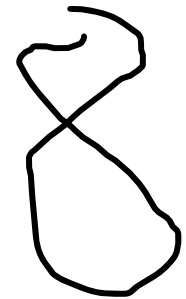
- For a given integer M
- First lowpass filter with cutoff frequency of π/M
- Then downsample by factor M
- Will avoid aliasing

No aliasing filter??

- Shouldn't really do this but...
 - Signal would already have to be sufficiently bandlimited
 - Equivalently, must be sufficiently oversampled
- Original signal bandlimited to Ω_N
- New sampling rate $\Omega'_S = \frac{\Omega_S}{M}$
- To satisfy Nyquist criterion:
 - $\Omega'_S \geq 2\Omega_N$
- Thus:
- $\Omega_S \geq 2M\Omega_N$

Problem for you...

- A continuous-time signal bandlimited to $\Omega_N = 500\pi$ rad/s
- The signal is sampled at $\Omega_S = 8000\pi$ rad/s
- The resulting discrete-time sequence is to be downsampled by an integer factor M , without using an anti-aliasing filter.
 - Determine the largest value of M that can be used without introducing aliasing.
 - For your value of M , what is the new sampling frequency?
 - Explain why choosing $M+1$ would cause aliasing.



Increase Sampling Rate

Increase sampling rate by integer factor

- Increase sampling rate of $x[n]$ by integer factor L
- Want
 - $x_i[n] = x_c(nT_i), \quad T_i = T/L$
- Where you can take the samples from
 - $x[n] = x_c(nT)$
- This is upsampling
 - $x_i[n] = x[n/L] = x_c(nT/L), \quad n = 0, \pm L, \pm 2L, \dots$
-

System

- Note distinction between $x_i[n]$ and $x_e[n]$
- $x_e[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - kL]$
- $x_e[n] = \begin{cases} x[n/L], & n = 0, \pm L, \pm 2L, \dots \\ 0, & \text{otherwise} \end{cases}$
- Expander
- LPF reconstructs the complete sequence

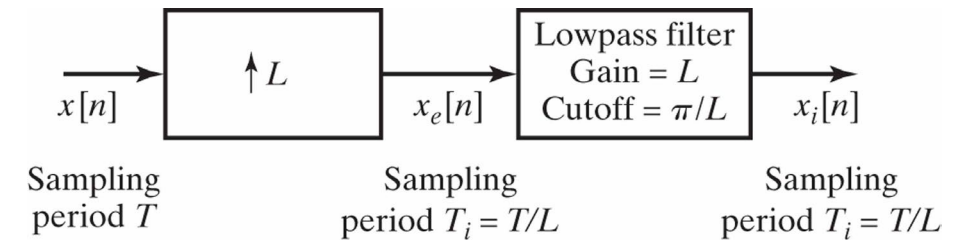
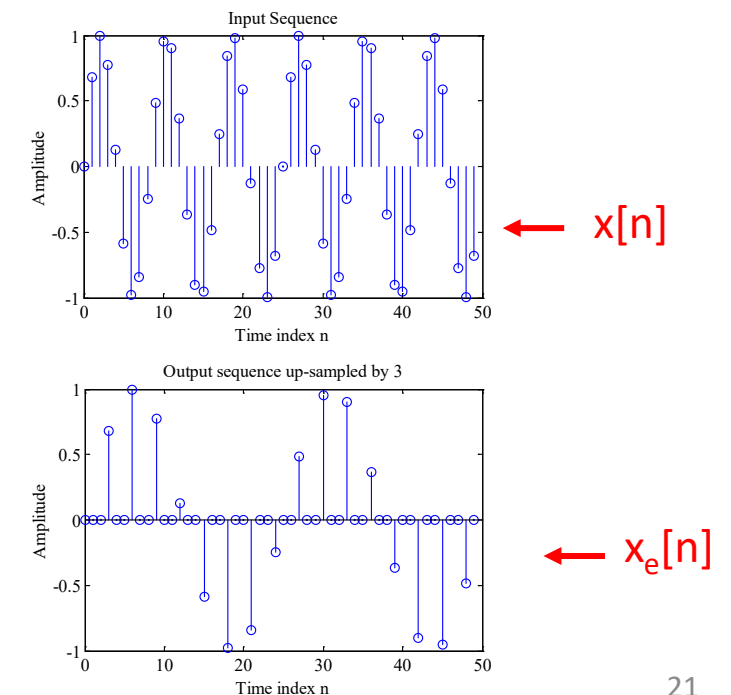


Figure 4.23 General system for sampling rate increase by L .



Increase sampling rate by integer factor

- Frequency domain interpretation, can express the FT as:
- $X_e(e^{j\omega}) = \sum_{n=-\infty}^{\infty} (\sum_{k=-\infty}^{\infty} x[k] \delta[n - kL]) e^{-j\omega n}$
- $= \sum_{k=-\infty}^{\infty} x[k] e^{-j\omega Lk} = X_e(e^{j\omega L})$
- So FT of output of expander is a frequency scaled version of FT of the input, $\omega \rightarrow \omega L$ with ω now normalised by
- $\omega = \Omega T_i$

Get the desired signal

- $X_i(e^{j\omega})$ can be obtained from $X_e(e^{j\omega})$ by correcting the amplitude scaling from $1/T$ to $1/T_i$ and removing all frequency-scaled images of $X_c(j\Omega)$ except at integer multiples of 2π
- This requires a lowpass filter, gain of L and cut-off frequency π/L

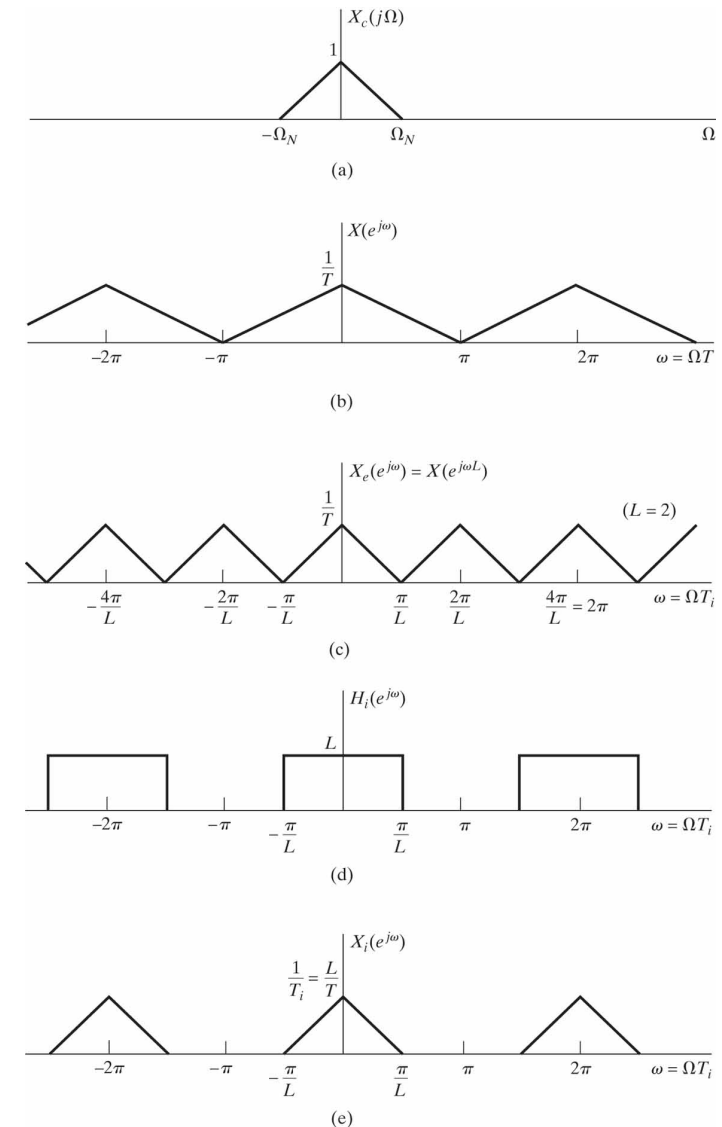


Figure 4.24 Frequency-domain illustration of interpolation.

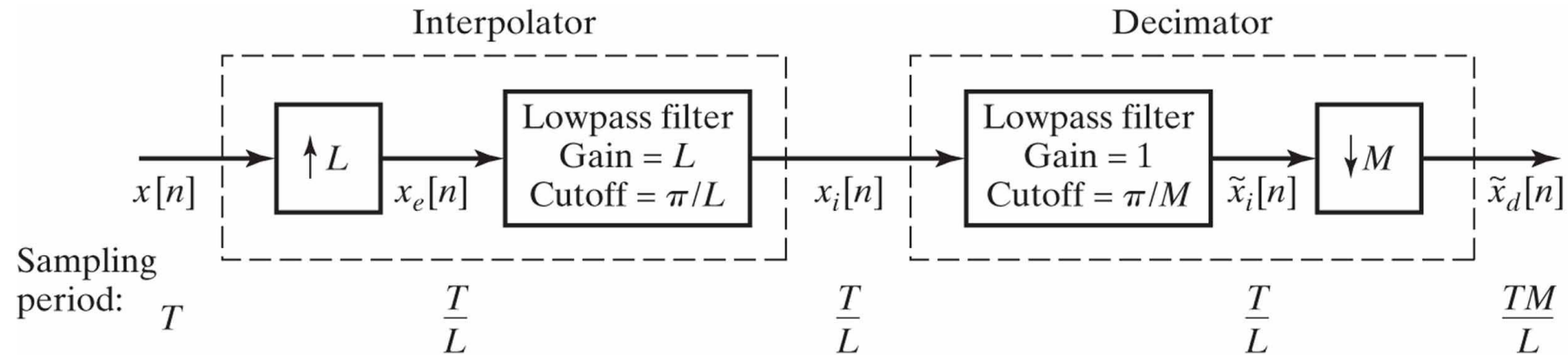
Interpolation

- Can view this as interpolation in the time domain
- Impulse response of the LPF is
 - $h_i[n] = \frac{\sin(\pi n/L)}{\pi n/L}$
- Can see that:
 - $x_i[n] = \sum_{k=-\infty}^{\infty} x[k] \frac{\sin(\pi(n-kL)/L)}{\pi(n-kL)/L}$
- Ideal case $h_i[0] = 1$ and $h_i[n] = 0 \quad n = \pm L, \pm 2L, \dots$
- Can show (you should try!) this gives:
 - $x_i[n] = x[n/L] = x_c[nT/L] = x_c(nT_i)$

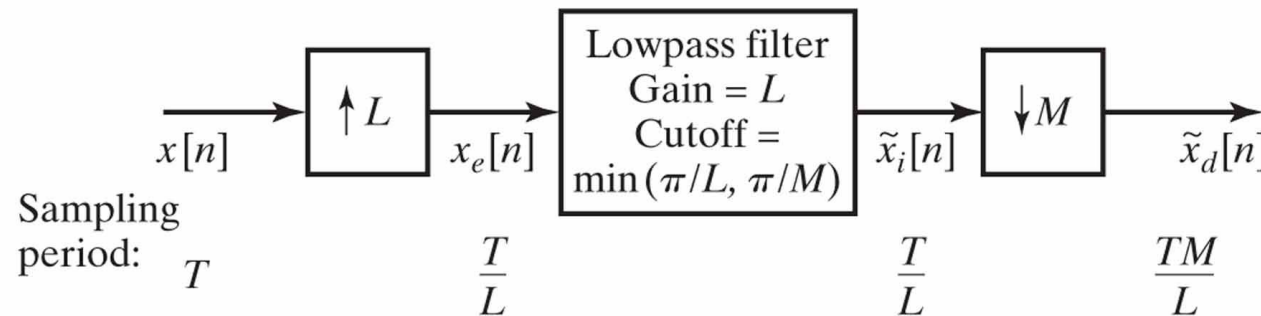
Practical Interpolation

- We looked at the ideal LPF
- Can only be approximated in reality
- Linear interpolation can be useful
 - Simplicity/accuracy trade-off

Change sample rate by non-integer factor



(a)



(b)

Combine decimation
and interpolation

Figure 4.29 (a) System for changing the sampling rate by a noninteger factor. (b) Simplified system in which the decimation and interpolation filters are combined.

Required Reading & other material

- Oppenheim & Schafer, Chapter 4, particularly 4.6
- If your DTFT knowledge is in need of sharpening:
 - <https://ocw.mit.edu/courses/res-6-007-signals-and-systems-spring-2011/resources/lecture-11-discrete-time-fourier-transform/>